

Aditya Saini

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EDUCATION

Purdue University

B.S. Computer Science (Systems Programming)

West Lafayette, IN

Graduation: May 2028 (Expected)

Relevant Coursework: Linear Algebra, Multivariate Calculus, Data Structures & Algorithms, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming

EXPERIENCE

Robotic Safety Research Assistant

PurSec, Purdue Computer Science

January 2026 – Present

West Lafayette, IN

- Researched machine learning safety for multimodal LLMs controlling robots, formulating clear attack hypotheses and evaluating model failure modes across 10+ AI2-THOR scenarios
- Built Python evaluation pipelines (Docker, AWS EC2) to run reproducible model regression tests, aggregate metrics with pandas/Matplotlib, and analyze inference-time safety verdicts
- Red-teamed vision/language baselines (GPT-4o, GPT-4o Vision) via a dual-brain setup (RoboGuard LTL verification + ASIMOV constitution checks), quantifying per-step and end-to-end failure rates
- Engineered multi-step physics attacks that chain individually-safe actions into unsafe outcomes, translating qualitative risks into concise, measurable evaluation criteria

Embedded Engineer

Purdue Electric Racing

Sep. 2025 – Present

West Lafayette, IN

- Wrote and debugged performance-critical C firmware for real-time sensor pipelines, profiling bottlenecks and optimizing command paths for production reliability
- Built telemetry and data-acquisition tooling to diagnose system behavior and ship iterative improvements under changing race-day requirements
- Owned I2C/SPI HAL APIs and UART dashboard drivers for high-frequency sensor I/O, collaborating across software and electrical teams

PROJECTS

Tava | Python, TypeScript, OpenAI, Firebase, React Native

September 2025

- Built AI-powered search and swipe-based matching to recommend collaborators for student project ideas, ranking candidates by shared interests via LLM-assisted retrieval.

Natural Language Shell | C, Python, OpenAI API, Git

June 2025 – January 2026

- Implemented a C/Python shell that maps natural-language queries to bash via local parsing and LLM inference, focusing on clear problem decomposition and end-to-end debugging.

FRC Team 2554 | Java, OpenCV, CUDA, Jetson Orin Nano

September 2021 – August 2025

- Led 40+ students as Build President, coordinating software/mechanical/electrical teams to District Championships after 18+ years.
- Built a CUDA-accelerated OpenCV vision pipeline for AprilTag detection and tuned PID/FF control, improving autonomous positioning accuracy by 66%.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, TypeScript, JavaScript

ML & Data: PyTorch, OpenCV, NumPy, pandas, Matplotlib, multimodal LLMs, model evaluation/inference, search & ranking

Systems & Tools: debugging, performance optimization, Git, Docker, AWS EC2, Linux, CUDA, Claude Code, Cursor

Foundations: Linear Algebra, Data Structures & Algorithms, critical problem formulation